WELLESLEY COLLEGE • DATA SCIENCE MAJOR CAPSTONE • Spring 2026

# Neighbourhood Socioeconomics & Food Inspection Outcomes in Boston, 2015–2025

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RESEARCH QUESTION

*How much do food safety inspection outcomes in Boston systematically differ across neighbourhoods of varying socioeconomic status? And has this gap widened over the past decade?*

**Yes, neighbourhood income is associated with food inspection outcomes in Boston, but the effect is small and it hasn't grown over the past decade.**

## i. Background & Motivation

- If neighbourhood disadvantage shapes pass/fail outcomes, then it could suggest **disparities in regulatory protection** across communities.
- Boston is a strong case-study because:
  - It documents day-to-day enforcement of health regulations across an entire city.
  - Its geocoded record covers >10 years and can be linked to census tracts and demographics.

→ Does Socio-Economic Status (SES) predict outcomes **after** accounting for establishment-level differences?

## ii. Data Sources

- **Boston Food Establishment Inspections** (2015–2025): 872,111 raw violation records, deduplicated to 100,211 unique establishment-visit inspections.
- **ACS 2019 5-year estimates**: census-tract level. Extracted: income, poverty, racial composition, foreign-born share.

→ Spatially joined to inspections via tract polygons.

| 91.6K | 6,912 | 174 | 46% |
|---|---|---|---|
| INSPECTIONS | ESTABLISHMENTS | CENSUS TRACTS | OVERALL FAIL RATE |

**Why is the fail rate so high (46%)?**

- Boston uses an **inclusive definition**: any inspection with at least one violation of any severity. (Most failures are minor, correctable issues, and not public health emergencies.)

## iii. Methods

Used **mixed-effects logistic regression** with random intercepts via establishments to avoid violating the independence assumption due to repeated inspections at certain establishments.

This also lets us decompose **within- vs. between-establishment** variation, which tells us whether a restaurant is the problem or whether the inspection/inspector is.

**Models fit:**

- **Null model:** how much variation comes from establishments themselves?
- **Income only:** how strong is the effect of income on its own?
- **Fully adjusted:** does income still matter once we control for business type, license age, inspection frequency, and time?
- **Income × time:** has the income gap widened over the past decade?

## iv. References

Inspectional Services Department, City of Boston. *Food Establishment Inspections.* Department of Innovation and Technology, data.boston.gov. Accessed April 3rd, 2026.

U.S. Census Bureau. *American Community Survey 5-Year Estimates*, 2015–2019. 2020, www.census.gov/programs-surveys/acs. Accessed April 3rd, 2026. Retrieved via the tidycensus R package.

## v. Exploratory Findings

From the EDA alone, we can already notice:

- A **weak negative trend** of inspection fail rate over time.
- Choropleth maps of pass rate vs. income show **visually flat** spatial overlap → any neighbourhood SES effect is likely **modest**.
- The gap between low- and high-income neighbourhoods has **persisted but not widened** over the decade.

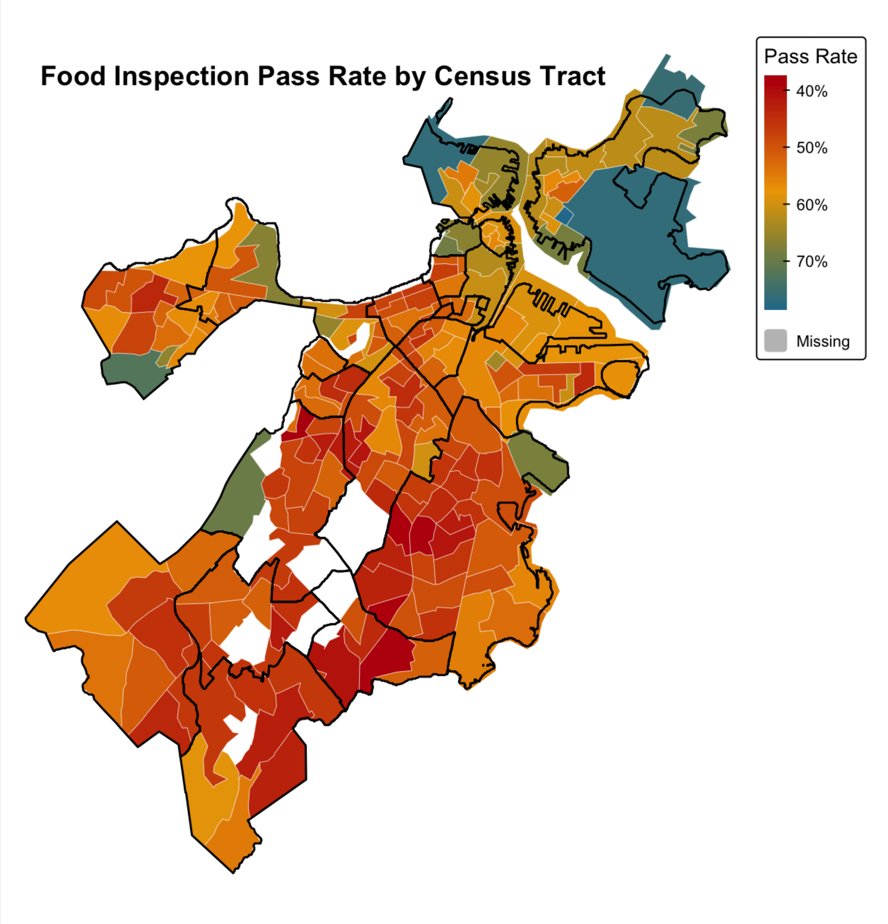


**Figure 1a.** *Inspection pass rate by census tract.*

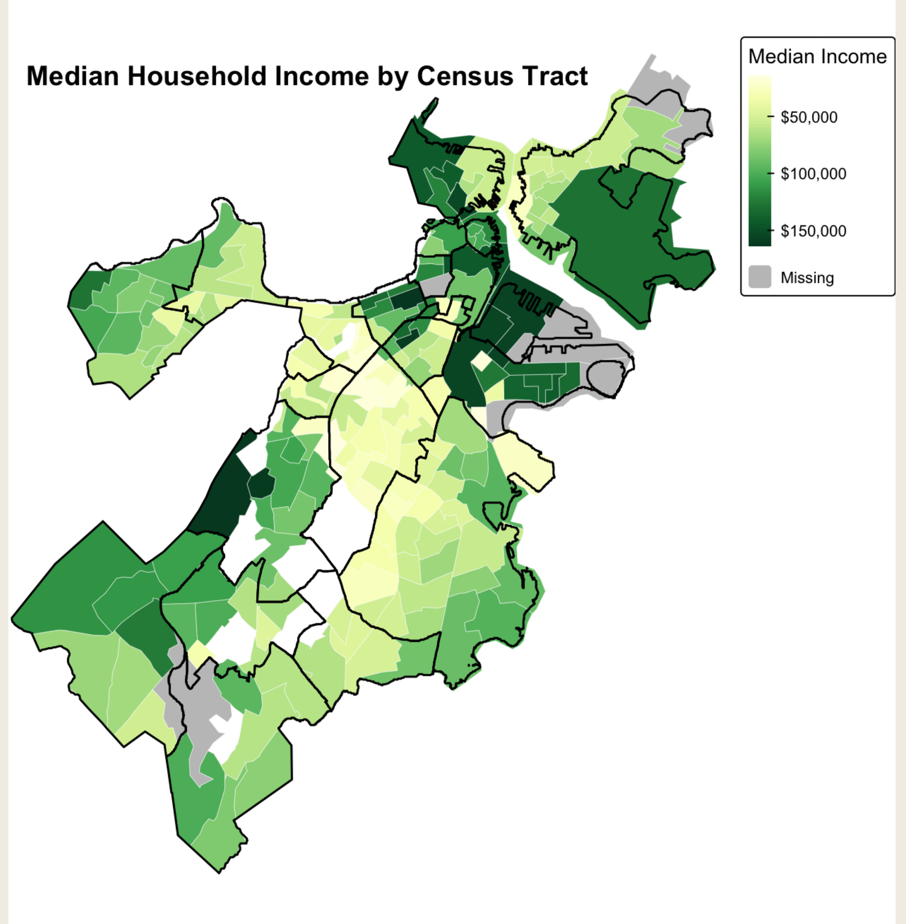


**Figure 1b.** *Median household income by census tract.*

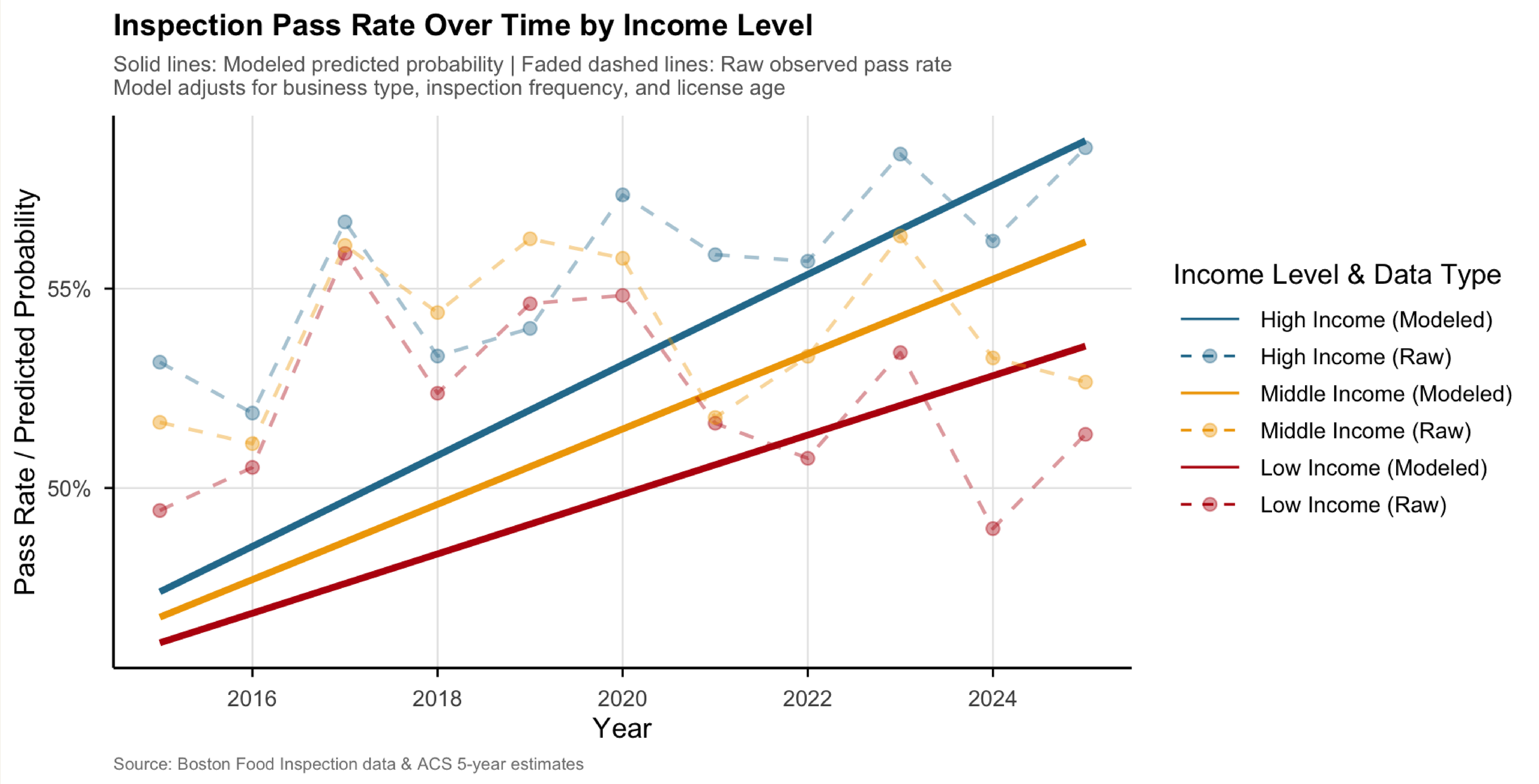


**Figure 2.** *Pass rate over time by income tercile. Dashed lines show raw observed rates; solid lines show modelled predicted probabilities adjusted for business type, inspection frequency, and license age. The lines run nearly parallel.*

## vi. Limitations & Data Ethics

**Limitations:**

- Only **one city** is analysed → limits generalisability to other urban contexts.
- ACS 2019 demographics are applied to a 10-year window, which may not capture neighbourhood demographic change.
- Violation **severity** is collapsed into a binary pass/fail.
- Inspection **frequency is endogenous** to outcomes (failures trigger re-inspections).

**Data ethics:**

- Neighbourhood-level patterns should not be read as statements about individuals.
- Findings should not be used to stigmatise specific neighbourhoods or operators.

## vii. Results & Discussion

- **Most of the variation isn't about neighbourhoods at all.** Only 4.6% of pass/fail variation comes from differences between establishments. The other 95% are the same restaurants passing some visits and failing others, which caps how many any neighbourhood factor can explain.
- The income **gap is real but small.** Moving from a low-income to a high-income tract shifts the predicted pass rate from 50.1% to 53.5%.
- This same gap has **not grown**. The income x time interaction is technically significant ($p = 0.006$) but the effect is tiny. We can see in Figure 2 that the lines stay almost parallel.
- Thus, what the establishments decide to do matters more than where they are located. Adding business type and inspection frequency improves the model much more than any other covariate. Mobile food establishments pass more often than sit-down restaurants, so does retail food.

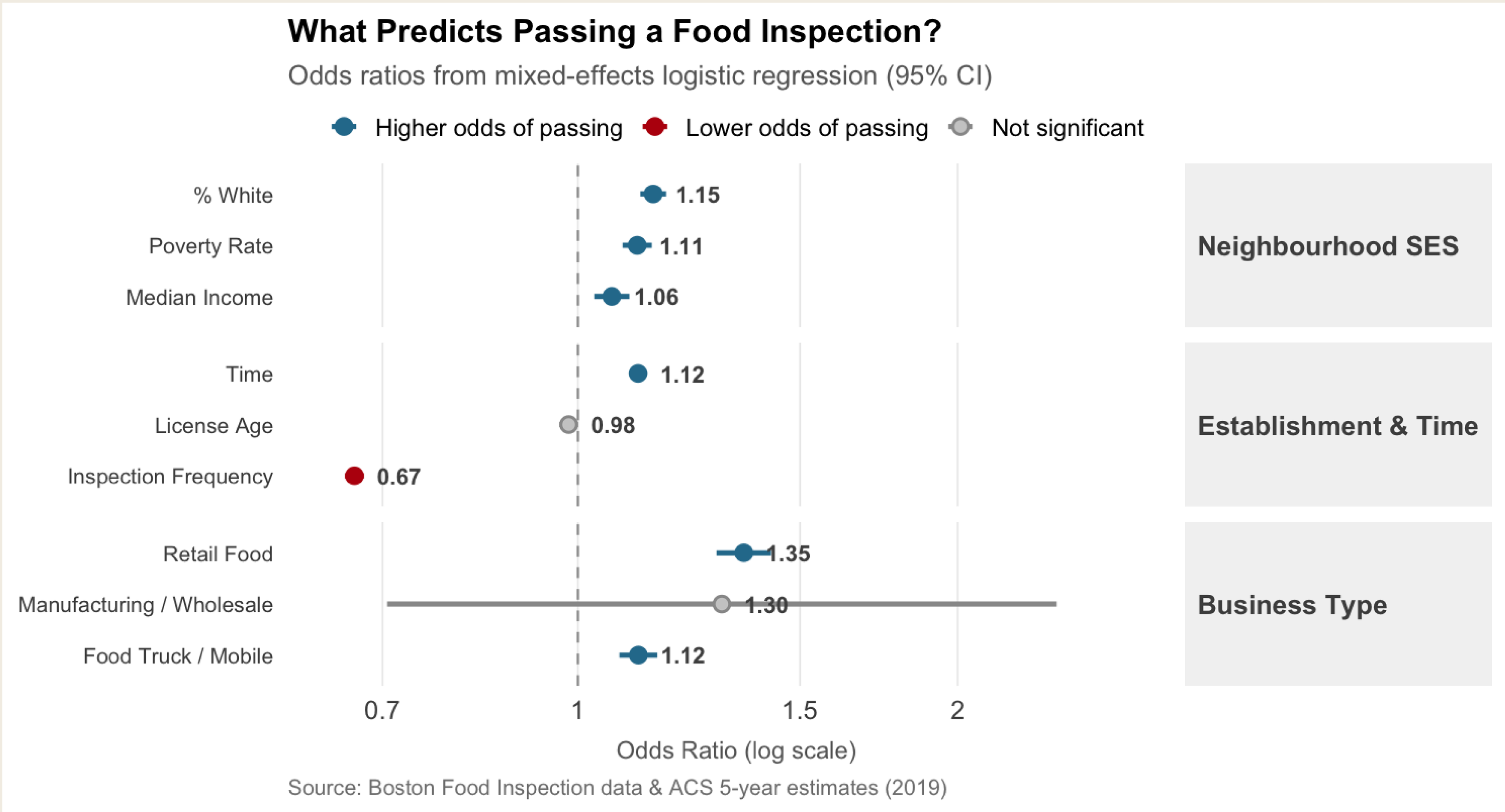


**Figure 3.** *Inspection frequency dominates the coefficient plot (OR ≈ 0.67). Median income is positive but modest (OR ≈ 1.06). Business type effects rival or exceed the income effect.*

***Model comparison (AIC):***

| *Model* | *AIC* | *ΔAIC* | *logLik* |
|---|---|---|---|
| *Null (intercept only)* | *99,978.5* | *0.0* | *−49,987.2* |
| *Income only* | *99,899.0* | *−79.4* | *−49,946.5* |
| *Poverty only* | *99,976.0* | *−2.4* | *−49,985.0* |
| *accentgold!20 Fully adjusted* | *97,248.4* | *−2,730.1* | *−48,613.2* |
| *Income × time* | *97,242.7* | *−2,735.8* | *−48,609.3* |

**Table 1.** *The fully adjusted model dominates simple SES-only specifications. Most of the AIC improvement comes from inspection frequency and business type — not demographics.*

***Two things to be careful about:***

- *Inspection frequency is circular: failing triggers re-inspections, which inflates the count. The strong negative coefficient is not just an effect.*
- *Sorting effect: differences across neighbourhoods may also reflect who opens where, not only how inspections are run.*

## viii. Conclusion

Neighbourhood income is associated with food inspection outcomes in Boston, but the effect is small and it hasn't grown over the past decade.

- The gap between low and high income tracts exists but is small.
- Differences in **business type** matter more than where a restaurant is located.
- Most of what drives pass/fail is *within-establishment noise*, not neighbourhood disadvantage.

In practice, this means that the data point toward establishment-level oversight rather than neighbourhood-level interventions. It also pushes back against the assumption that low-income areas of Boston are systematically failing inspections at much higher rates (they aren't!). → However, this analysis can't say whether more targeted oversight would actually *improve* outcomes.

**Policy implication:**

- Supports **establishment-level** rather than area-based regulatory attention.
- Pushes back against narratives that low-income Boston neighbourhoods fail inspections at dramatically higher rates.